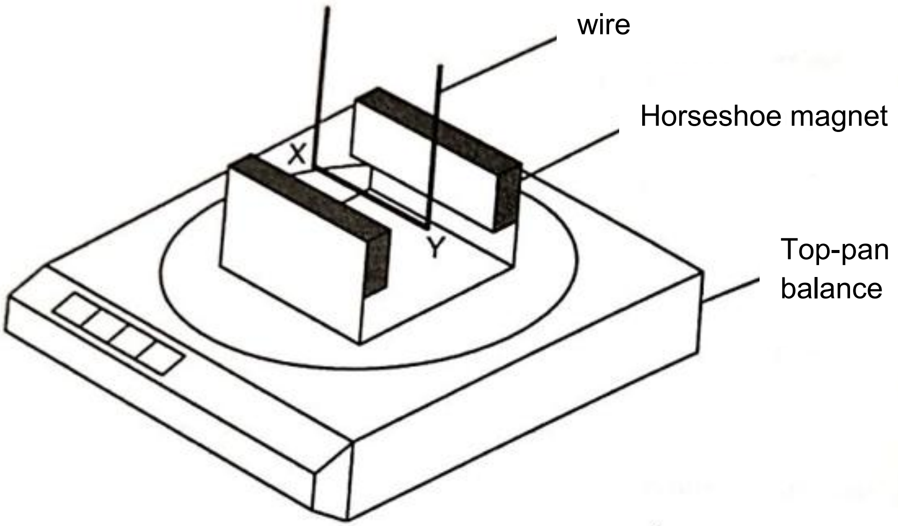
25. A horseshoe magnet rests on a top-pan balance with a wire situated between the poles of the magnet.



With no current in the wire, the reading on the balance is 142.0 g.

With a current of 2.0 A in the wire in the direction XY, the reading on the balance changes to 144.6 g.

What is the reading on the balance when there is a current of 3.0 A in the wire in the direction YX?

A 138.1 g **B** 140.7 g **C** 145.9 g **D** 148.5 g

Answer: A

Force on XY is equal and opposite to the force exerted on magnet (Newton's 3rd law)
Force on XY, $F = BIL \Rightarrow F \propto I$

Without current, balance is measuring mass of magnet which is 142.0 g.
With 2.0 A in XY, force on magnet causes an increase in mass $= 144.6 - 142.0 = 2.6$ g

With 3.0 A in YX, force on magnet causes a decrease in mass $= 142.0 - \frac{2.6}{2} \times 3 = 138.1$ g